Supplementary Material for "Distribution-Preserving Quantized Diffusion Models for Probabilistic Time Series Forecasting"

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This document complements the main paper with complete proofs of the theoretical results (Section S1), implementation details of DPQ-Diff and of the compared quantizers (Section S2), and additional results (Section S3). Numbers without the prefix S, such as Eq. (9) or Table I, refer to the main paper, and the code repository contains a script for every reported number.

S1. Proofs

Throughout this section, CRPS_S denotes the ensemble estimator of Eq. (2), $J = HN$ is the number of coordinates of a forecast window, and all statements about $\overline{\mathrm{CRPS}}$ hold window by window and therefore also for the average over the test windows.

A. Proof of Lemma 1 (Stability of the CRPS)

Ensemble statement. Let $x_{1:S}$ and $\hat{x}_{1:S}$ be two ensembles and o an observation, and write $u_s = |\hat{x}_s - x_s|$. By the reverse triangle inequality,

$$\big||\hat{x}_s - o| - |x_s - o|\big| \le u_s, \qquad \big||\hat{x}_s - \hat{x}_{s'}| - |x_s - x_{s'}|\big| \le u_s + u_{s'}. \tag{S1}$$

The difference of the two estimators splits into the difference of the accuracy terms and the difference of the spread terms. The first difference is bounded by $\frac{1}{S}\sum_s u_s$, and the second by

$$\frac{1}{2S^2}\sum_{s,s'=1}^{S}(u_s + u_{s'}) = \frac{1}{2S^2}\cdot 2S\sum_{s=1}^{S} u_s = \frac{1}{S}\sum_{s=1}^{S} u_s. \tag{S2}$$

Adding both bounds yields Eq. (8). The bound holds for every pairing of the members of both ensembles, and in particular for the pairing induced by shared sampling noise that we use throughout the paper.

Population statement. Let $(Z, \hat{Z})$ be an optimal coupling of P and Q for the Wasserstein-1 distance, and let $(Z', \hat{Z}')$ be an independent copy. By Eq. (1), the difference $\Delta = \mathrm{CRPS}(Q, o) - \mathrm{CRPS}(P, o)$ satisfies

$$\begin{aligned}|\Delta| &\le \mathbb{E}\big|\,|\hat{Z} - o| - |Z - o|\,\big| + \tfrac{1}{2}\mathbb{E}\big|\,|\hat{Z} - \hat{Z}'| - |Z - Z'|\,\big| \\ &\le \tfrac{3}{2}\mathbb{E}|\hat{Z} - Z| + \tfrac{1}{2}\mathbb{E}|\hat{Z}' - Z'|,\end{aligned} \tag{S3}$$

which equals $2\,W_1(P, Q)$ because both couplings have the same distribution. □

B. Proof of Theorem 1 (Location–Scale–Shape Decomposition)

Decomposition. Under shared endpoint and ancestral noises, the s-th quantized and full-precision sample paths are $\widehat{\mathbf{Y}}_q^{(s)} = \hat{f} + \sqrt{\hat{g}} \odot \hat{Z}_0^{(s)}$ and $\widehat{\mathbf{Y}}^{(s)} = f + \sqrt{g} \odot Z_0^{(s)}$. Adding and subtracting $\sqrt{g} \odot \hat{Z}_0^{(s)}$ gives

$$\begin{aligned}\widehat{\mathbf{Y}}_q^{(s)} - \widehat{\mathbf{Y}}^{(s)} = (\hat{f} - f) + (\sqrt{\hat{g}} - \sqrt{g}) \odot \hat{Z}_0^{(s)} \\ + \sqrt{g} \odot (\hat{Z}_0^{(s)} - Z_0^{(s)}).\end{aligned} \tag{S4}$$

For a coordinate j, the triangle inequality bounds the absolute value of the j-th entry by $|\hat{f}_j - f_j| + |\sqrt{\hat{g}_j} - \sqrt{g_j}|\,|\hat{Z}_{0,j}^{(s)}| + \sqrt{g_j}\,|\hat{Z}_{0,j}^{(s)} - Z_{0,j}^{(s)}|$. Averaging over s and applying Eq. (8) to coordinate j yields

$$\begin{aligned}&\mathrm{CRPS}_S\big(\widehat{\mathbf{Y}}_{q,j}, o_j\big) - \mathrm{CRPS}_S\big(\widehat{\mathbf{Y}}_j, o_j\big) \\ &\quad\le 2\big(|\hat{f}_j - f_j| + |\sqrt{\hat{g}_j} - \sqrt{g_j}|\,\hat{m}_j + \sqrt{g_j}\,\bar{\delta}_j\big),\end{aligned} \tag{S5}$$

and averaging over the J coordinates proves Eq. (9).

Accumulation along the reverse chain. Write the full-precision and the quantized reverse steps as $Z_{t-1} = a_t Z_t - b_t \epsilon_\theta(Z_t, t) + \sigma_t \xi_t$ and $\hat{Z}_{t-1} = a_t \hat{Z}_t - b_t \hat{\epsilon}_\theta(\hat{Z}_t, t) + \sigma_t \xi_t$, where the conditioning inputs of both denoisers are absorbed into their definitions, so that $e_t = \hat{\epsilon}_\theta(\hat{Z}_t, t) - \epsilon_\theta(\hat{Z}_t, t)$ also contains the effect of the quantized backbones on the denoiser input. With $d_t = \hat{Z}_t - Z_t$ and the shared endpoint $Z_T = \hat{Z}_T$, we have $d_T = \mathbf{0}$ and

$$d_{t-1} = a_t d_t - b_t\big(e_t + \epsilon_\theta(\hat{Z}_t, t) - \epsilon_\theta(Z_t, t)\big). \tag{S6}$$

The Lipschitz continuity of $\epsilon_\theta(\cdot, t)$ yields $\|d_{t-1}\|_2 \le (a_t + b_t L_t)\|d_t\|_2 + b_t\|e_t\|_2$. By induction over $t = T, T-1, \ldots, 1$, this recursion gives

$$\|d_{t-1}\|_2 \le \sum_{u=t}^{T} b_u \|e_u\|_2 \prod_{v=t}^{u-1}(a_v + b_v L_v), \tag{S7}$$

and setting $t = 1$ proves Eq. (10) with $\Gamma_u = b_u \prod_{v=1}^{u-1}(a_v + b_v L_v)$. The weights Γ_u grow with u whenever $a_v + b_v L_v > 1$, so that errors made early in the reverse chain, that is, at large t, pass through more contracting or expanding steps before they reach the forecast. □

C. Proof of Proposition 1 (Step Groups)

Bound. Inserting $\|e_t\|_2 \le c\, r_{\gamma(t)}$ into Eq. (10) gives $\|\hat{Z}_0 - Z_0\|_2 \le c\sum_t \Gamma_t r_{\gamma(t)} = c\sum_k r_k \sum_{t \in \mathcal{T}_k} \Gamma_t = c\,\Phi$.

Monotonicity under refinement. Let a group $\mathcal{T}_k$ be split into $\mathcal{T}_k' \cup \mathcal{T}_k''$. Both new ranges are maxima over subsets

of $\mathcal{T}_k$ and therefore at most r_k, so that the contribution $r_k \sum_{t \in \mathcal{T}_k} \Gamma_t$ cannot increase, since all $\Gamma_t \geq 0$. By induction, every refinement of a partition, and in particular the partition into single steps, attains a value of Φ that is at most that of the coarser partition.

Optimality of contiguous groups. Let $\mathcal{P}^\star$ be an optimal partition into G groups with ranges $r_1^\star < \cdots < r_{G'}^\star$, where $G' \leq G$ counts the distinct values. Assign every step t to the group with the smallest range $r_k^\star \geq m_t$; such a group exists because the group that contains t in $\mathcal{P}^\star$ has a range of at least m_t. Each group keeps the step that attains its maximum, so that all ranges remain unchanged, and no step is assigned a larger range than before, so that Φ does not increase and the new partition is also optimal. In the new partition, group k contains exactly the steps with $r_{k-1}^\star < m_t \leq r_k^\star$. If m_t is monotone in t, these level sets are contiguous runs of steps, and splitting runs, if $G' < G$, does not increase Φ by the previous argument. □

When m_t is only approximately monotone, as in the measured trajectories of Fig. S1, the same argument shows that every partition can be converted into one whose groups are level sets of m_t, so that contiguous runs remain optimal up to the steps at which the monotonicity is violated.

D. Anchors Control the Location and Scale Terms

The fine-tuning objective of Eq. (14) penalizes the squared deviations of $\hat{f}$ and of $\log \hat{g}$. The following corollary makes precise how these penalties bound the first two terms of Eq. (9).

Corollary S1 (Anchor bounds)**.** *Let $\Delta = \log \hat{g} - \log g$ with $|\Delta_j| \leq \kappa$ for all coordinates. Then*

$$\tfrac{1}{J}\|\hat{f} - f\|_1 \leq \tfrac{1}{\sqrt{J}}\|\hat{f} - f\|_2, \tag{S8}$$

$$\tfrac{1}{J}\big\langle |\sqrt{\hat{g}} - \sqrt{g}|, \hat{m}\big\rangle \leq \tfrac{e^{\kappa/2}}{2J}\left\|\sqrt{g} \odot \hat{m}\right\|_2 \|\Delta\|_2. \tag{S9}$$

Proof. The first inequality is the Cauchy–Schwarz inequality $\sum_j |v_j| \leq \sqrt{J}\,\|v\|_2$. For the second, $\sqrt{\hat{g}_j} - \sqrt{g_j} = \sqrt{g_j}\,(e^{\Delta_j/2} - 1)$, and the mean value theorem gives $|e^u - 1| \leq |u|\, e^{|u|}$ for every real u, so that $|\sqrt{\hat{g}_j} - \sqrt{g_j}| \leq \tfrac{1}{2} e^{\kappa/2} \sqrt{g_j}\,|\Delta_j|$. Multiplying by $\hat{m}_j$, summing over j, and applying the Cauchy–Schwarz inequality to the vectors $\sqrt{g} \odot \hat{m}$ and Δ completes the proof. □

Hence the location and log-variance anchors bound the location and scale terms of Theorem 1 up to factors that depend only on the full-precision forecaster, and they do so for every sample path at once. The noise anchor penalizes the per-step error e_t on states drawn from the forward process, which share the marginal distribution of the reverse states of a well-trained forecaster, and thereby controls the inputs of the accumulation bound in Eq. (10).

S2. Implementation Details

A. Forecasters and Data

We train DiffPTS and NsDiff with their official implementations and the configurations released with them, using a history of $L = 168$ steps, $S = 100$ sample paths, and the train, validation, and test splits of the NsDiff and DiffPTS protocol. DiffPTS uses $T = 20$ reverse steps with a linear schedule, and NsDiff uses its uncertainty-aware schedule with $T = 100$ steps. Each quantized model is derived from, and compared with, a specific trained FP32 forecaster, and the quantized and full-precision forecasters are evaluated on identical test windows with identical sampling noise, which isolates the effect of quantization from the randomness of the sampler.

B. Quantizers and Storage Accounting

All quantizers are symmetric and uniform as in Eq. (6). Weights use one scale per output channel, initialized to the maximum absolute value of the channel, and activations use static per-tensor scales; the denoiser activations additionally receive one scale per step group. Every linear and convolutional layer of f_ϕ, g_ψ, and ϵ_θ is quantized, including embeddings, attention projections, feed-forward layers, and output heads, whereas biases, normalization parameters, and the step embeddings of the denoiser remain in floating point. The storage of a quantized model counts the packed integer weights with one half-precision scale per output channel and all remaining parameters and buffers in single precision, and the FP32 reference stores all parameters in single precision. Deterministic positional encodings are recomputed at load time and are therefore stored by neither model. Bit operations count $b_w \cdot b_a$ for every MAC of a quantized layer and $32 \cdot 32$ for every MAC that stays in floating point.

C. Step-Aware Calibration

The calibration windows are drawn from the validation split. For every window, the FP32 forecaster runs its reverse process with $S_{\mathrm{cal}} = 100$ sample paths, and the inputs of every denoiser layer are recorded at every reverse step. The reverse steps are partitioned into $G = 5$ contiguous groups of equal length, and for every layer and group the clipping ratio is selected from a grid of 36 values between 0.3 and 1 by Eq. (12). The backbones receive one range per input tensor, estimated on the same windows with the same error-minimizing search. The lookup table is computed once per forecaster and stored with its integer weights.

D. Output-Preserving Rounding

The rounding variables $\mathbf{V}$ of Eq. (11) are initialized such that $h(\mathbf{V})$ equals the fractional part of $\mathbf{W}/\mathbf{s}$, which reproduces the full-precision weights at the start, and they are optimized with Adam on mini-batches of recorded layer inputs. The regularizer is disabled during the first fifth of the iterations and then enabled with an exponent β that is annealed linearly from 20 to 2, after which every entry of $h(\mathbf{V})$ is rounded to a hard decision. Backbone layers are fitted to their inputs on the calibration windows, and denoiser layers to their inputs along the recorded reverse trajectories, so that the rounding of every layer is adapted to the step-dependent inputs that it receives at inference time.

E. Distribution-Preserving Fine-Tuning

Fine-tuning starts from the calibrated post-training pipeline, keeps full-precision master weights, and quantizes them per output channel in every forward pass, with gradients passed through the rounding by the straight-through estimator. The activation scales remain fixed at the calibrated lookup table, and every activation is quantized with the scale of the step group of its sample. We use Adam with a small learning rate, the anchor weight $\lambda = 1$, antithetic pairs of reverse steps as in DiffPTS, and a few epochs of at most 300 mini-batches. After every epoch, the validation CRPS of the quantized forecaster is computed with the deployed sampler, and the calibrated starting point is scored in the same way, so that the deployed forecaster is the best pipeline observed on the validation windows.

F. Compared Quantizers

PTQ4DM [1] draws calibration steps from a normal distribution centred on the middle of the reverse process, estimates one range per activation tensor with an error-minimizing clipping search, and rounds weights to the nearest grid point. Q-Diffusion [2] calibrates on uniformly sampled reverse steps with the same range search and learns the rounding of every layer by output reconstruction with the same optimizer, iterations, and regularizer as DPQ-Diff. Both quantizers use per-output-channel weight scales, and we run them with the calibration windows and the number of sample paths of DPQ-Diff, both on the noise-prediction network only and on the full pipeline, so that the full-pipeline variants have exactly the bit-widths and the model size of DPQ-Diff.

G. Hardware and Reproducibility

All experiments run on a single NVIDIA RTX 3090 with 24 GB of memory. Integer arithmetic is emulated with fake quantization in PyTorch, which reproduces the outputs of an integer implementation up to floating-point summation order. The repository provides the implementation of DPQ-Diff on top of the official DiffPTS code base together with a script that trains a forecaster and quantizes it with the default configuration on any of the benchmarks.

S3. Additional Results

A. Run-Level Comparison with the Baselines

Table S1 compares DPQ-Diff with the full-pipeline variants of Q-Diffusion and PTQ4DM run by run, pairing the runs that quantize the same FP32 forecaster. With 4-bit weights, DPQ-Diff attains a lower CRPS than Q-Diffusion in 21 of the 27 runs and than PTQ4DM in 26 runs, and the one-sided Wilcoxon signed-rank test confirms both advantages with $p < 10^{-4}$. With 8-bit weights, where all quantizers are close to the FP32 forecaster, DPQ-Diff still wins the majority of runs against both baselines, and both differences are significant at the 5% level.

TABLE S1
RUN-LEVEL COMPARISON OVER THE NINE BENCHMARKS AND THREE RUNS (27 PAIRED RUNS PER SETTING): NUMBER OF RUNS IN WHICH DPQ-DIFF ATTAINS A LOWER CRPS THAN THE BASELINE, MEAN DIFFERENCE OF THE RELATIVE CRPS CHANGE IN PERCENTAGE POINTS, AND ONE-SIDED WILCOXON SIGNED-RANK p-VALUE.

Setting	Baseline	Wins	Mean diff.	p-value
W4A8	Q-Diffusion	21/27	-2.13	$2.0 \cdot 10^{-5}$
W4A8	PTQ4DM	26/27	-3.59	$5.2 \cdot 10^{-8}$
W8A8	Q-Diffusion	17/27	-1.17	$3.2 \cdot 10^{-3}$
W8A8	PTQ4DM	17/27	-1.14	$1.5 \cdot 10^{-2}$

TABLE S2
CALIBRATION AND SHARPNESS OF THE QUANTIZED FORECASTERS: COVERAGE OF THE CENTRAL 90% INTERVAL AND ITS MEAN WIDTH FOR THE FP32 FORECASTER, AND THE CHANGE IN COVERAGE (PERCENTAGE POINTS) AND IN WIDTH (%) OF DPQ-DIFF (MEAN OF THREE RUNS).

	FP32		DPQ-Diff W8A8		DPQ-Diff W4A8	
Dataset	Cov.	Width	ΔCov.	ΔWidth	ΔCov.	ΔWidth
ETTh1	95.6	2.67	+0.26	−0.97	+0.13	−1.44
ETTh2	93.0	2.60	+0.17	−0.15	−0.02	−0.68
ETTm1	90.1	1.87	+0.05	−0.01	−0.16	+0.36
ETTm2	96.5	2.33	+0.04	−1.35	+0.05	−0.46
ECL	85.2	1.06	−0.01	+0.12	+0.19	+0.83
Exchange	85.3	1.64	+1.66	−0.21	+1.11	−0.88
ILI	74.0	2.26	−0.00	−3.87	+1.79	−4.09
Solar	80.7	0.90	+0.68	+0.03	−0.19	+1.46
Traffic	92.0	1.71	−0.11	−0.05	−0.02	−0.06
Mean $\lvert\Delta\rvert$			0.33	0.75	0.41	1.14

B. Calibration and Sharpness

Table S2 reports the coverage of the central 90% interval and the mean interval width for every benchmark. The coverage of DPQ-Diff deviates from that of the FP32 forecaster by only 0.33 percentage points at W8A8 and by 0.41 percentage points at W4A8 on average, while the intervals become narrower on 7 and 6 of the nine benchmarks, respectively. The quantized forecasters are therefore as well calibrated as the FP32 forecaster and slightly sharper, which is consistent with their lower CRPS.

C. Fine-Tuning the Full-Precision Forecaster

Table S3 fine-tunes the FP32 forecaster with the objective, data, and budget of DPQ-Diff on the six smaller benchmarks, on which the quantized forecasters improve most. Additional training of the FP32 forecaster improves its CRPS on some benchmarks but degrades it by up to +3.07% on others and raises the MSE on average, whereas the quantized forecasters of DPQ-Diff improve the CRPS on average by a larger margin and stay within +0.22% of the original forecaster on every benchmark. The quantization noise and the anchors to the original forecaster thus regularize fine-tuning, which explains why the quantized forecasters can surpass the model from which they were derived.

TABLE S3
FINE-TUNING THE FP32 FORECASTER WITH THE OBJECTIVE, DATA, AND BUDGET OF DPQ-DIFF VERSUS THE QUANTIZED FORECASTERS OF DPQ-DIFF: CRPS AND MSE CHANGE (%) RELATIVE TO THE ORIGINAL FP32 FORECASTER (MEAN OF THREE RUNS). BOLD: BETTER THAN THE FINE-TUNED FP32 FORECASTER.

	FP32, fine-tuned		DPQ-Diff W8A8	DPQ-Diff W4A8	
Dataset	ΔCRPS	ΔMSE	ΔCRPS	ΔCRPS	ΔMSE
ETTh1	−3.40	−2.57	−1.36	−1.82	−2.14
ETTh2	+2.71	+9.55	**−0.86**	**−0.16**	**−0.27**
ETTm1	−0.01	−0.84	**−0.09**	+0.22	−0.03
ETTm2	+3.07	+11.56	**−0.62**	**+0.02**	**−0.26**
Exchange	−2.80	−7.19	**−3.58**	**−4.56**	**−10.13**
ILI	−5.95	−8.27	−1.80	**−6.06**	**−9.89**
Mean	−1.07	+0.37	**−1.38**	**−2.06**	**−3.79**
Worst	+3.07	+11.56	**−0.09**	**+0.22**	**−0.03**

TABLE S4
FINAL DEVIATION $\frac{1}{J}\|\hat{Z}_0 - Z_0\|_1$ OF THE STANDARDIZED SAMPLE PATHS FROM THE FP32 TRAJECTORY WITH 4-BIT DENOISER ACTIVATIONS AND MAXIMUM RANGES (MEAN OVER RUNS WITH SHARED SAMPLING NOISE), FOR ONE RANGE PER TENSOR ($G = 1$) AND FOR THE STEP GROUPS OF DPQ-DIFF ($G = 5$).

Dataset	Runs	$G = 1$	$G = 5$ (DPQ-Diff)	Reduction
ETTh1	3	0.0069	**0.0059**	−14%
ETTm1	3	0.2006	**0.1655**	−18%
Exchange	3	0.0101	**0.0069**	−32%
ILI	3	0.0080	**0.0063**	−22%
Solar	3	0.0218	**0.0170**	−22%
Mean				−22%

D. Shape Error along the Reverse Chain

Table S4 complements Fig. 4(c) with the final deviation of the standardized sample paths for every measured benchmark. Step groups reduce the deviation on every benchmark, by 14–32% and by 22% on average, which confirms principle (P2) and Proposition 1 on the actual reverse chains of the forecaster.

E. Range Drift on All Benchmarks

Figure S1 repeats the range analysis of Fig. 5(a) for every benchmark. The drift is most pronounced on ETTh2, ETTm1, ETTm2, ECL, Solar, and Traffic, where the ranges of the deeper denoiser layers decrease almost monotonically by a factor of two to five along the reverse trajectory. This is precisely the regime in which Proposition 1 identifies contiguous step groups as optimal, and the five groups of DPQ-Diff track the profile closely with one range per group.

F. Calibration Data

DPQ-Diff calibrates on the validation windows of a benchmark, which comprise between one and 32 windows under the protocol of NsDiff and DiffPTS. Figure S2 calibrates the post-training stage on 2, 4, and 16 windows spaced evenly over the validation split of the four ETT benchmarks and compares it with the full split. Four windows already match the full calibration set: averaged over the four benchmarks, the CRPS changes by −0.24% and +0.12% relative to the FP32 forecaster at W8A8 and W4A8, compared with +0.02% and +0.31% with the full split, and on every benchmark and bit-width four windows come within 0.01 percentage points of the full split or improve on it. A forecaster can therefore be quantized as soon as a few windows of recent history are available, for example right after it has been retrained on fresh data.

TABLE S5
NOTATION USED THROUGHOUT THE PAPER.

Symbol	Meaning
$\mathbf{X} \in \mathbb{R}^{L\times N}$, $\mathbf{Y} \in \mathbb{R}^{H\times N}$	history and future window of N variables
f_ϕ, g_ψ, ϵ_θ	mean backbone, variance backbone, denoiser
S, T	sample paths per forecast, reverse steps
Z_t, $\widehat{\mathbf{Y}}$	standardized reverse state, sample path
$\overline{\mathrm{CRPS}}$, J	mean CRPS over the $J = HN$ coordinates
$Q_b(\cdot; s)$, q_b	b-bit quantizer with scale s, integer bound
WxAy	x-bit weights and y-bit activations
G, $\mathcal{T}_k$, $\gamma(t)$	step groups, group k, group of step t
$\mathcal{L}_{\mathrm{ELBO}}$, $\mathcal{L}_{\mathrm{anc}}$, λ	ELBO, anchor loss, anchor weight
MACs, BOPs	multiply–accumulates, bit operations

TABLE S6
MODEL STATISTICS AND FOOTPRINT OF DIFFPTS PER BENCHMARK: VARIABLES N, PARAMETERS, MACS PER FORECAST (S=100, T=20) WITH THE SHARE OF THE DENOISER, AND STORAGE OF THE FP32 FORECASTER, OF THE NOISE-NETWORK QUANTIZERS, AND OF DPQ-DIFF.

		Params	GMACs	ϵ_θ	FP32	Noise net	DPQ-Diff (MB)	
Dataset	N	(M)		(%)	(MB)	(MB)	W8A8	W4A8
ETTh1	7	6.77	15.2	95.8	27.1	27.0	6.86	3.48
ETTh2	7	6.77	15.2	95.8	27.1	27.0	6.86	3.48
ETTm1	7	6.77	15.2	95.8	27.1	27.0	6.86	3.48
ETTm2	7	6.77	15.2	95.8	27.1	27.0	6.86	3.48
ECL	321	9.35	103.8	98.1	37.4	36.4	9.48	4.82
Exchange	8	7.91	16.4	90.7	31.6	31.5	8.04	4.10
ILI	7	7.83	3.8	70.6	31.3	31.2	7.95	4.06
Solar	137	8.50	52.4	96.8	34.0	33.5	8.63	4.40
Traffic	862	11.82	254.9	99.0	47.3	45.0	11.97	6.07

S4. NOTATION AND MODEL STATISTICS

Table S5 summarizes the notation of the main paper, and Table S6 lists the number of variables, the parameters and MACs of every DiffPTS forecaster, the share of the denoiser in the arithmetic, and the storage of the FP32 forecaster, of the forecaster whose noise network alone is quantized, and of the forecasters of DPQ-Diff. The noise-network-only quantizers leave the size of every forecaster almost unchanged, whereas DPQ-Diff compresses each of them by 3.9× and 7.8×, which is the memory–compute asymmetry of Observation 1 at the level of individual benchmarks.

REFERENCES

[1] Y. Shang, Z. Yuan, B. Xie, B. Wu, and Y. Yan, "Post-training quantization on diffusion models," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 1972–1981.

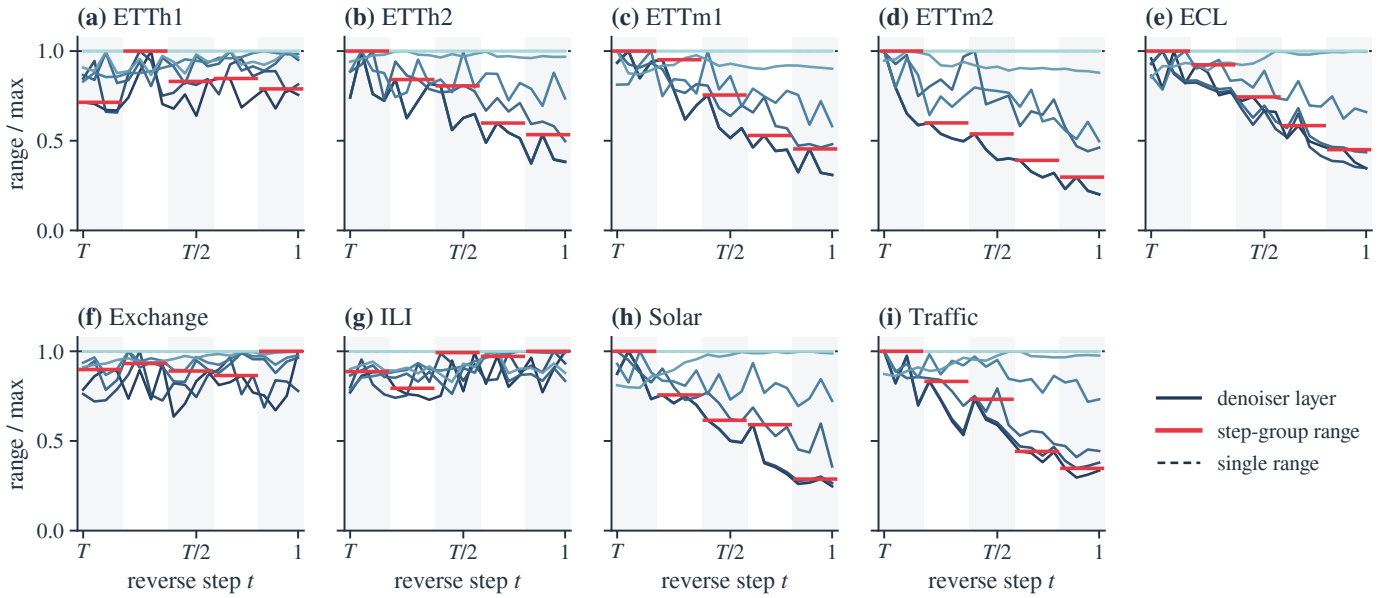


Fig. S1. Input range of every denoiser layer along the reverse trajectory for all nine benchmarks (first run), normalized per layer; red segments mark the range of each of the five step groups for the layer with the largest drift.

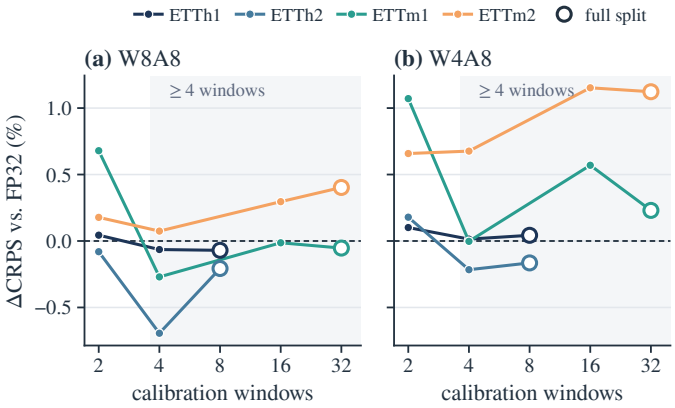


Fig. S2. CRPS change of the post-training stage of DPQ-Diff relative to the FP32 forecaster when calibrated on 2, 4, and 16 validation windows and on the full validation split (hollow markers), mean of three runs.

[2] X. Li, Y. Liu, L. Lian, H. Yang, Z. Dong, D. Kang, S. Zhang, and K. Keutzer, "Q-Diffusion: Quantizing diffusion models," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 17 489–17 499.